

Direct Two-Corner Primitive Planes in a Fixed-Shift Missing-Native Gram:

Exact Entry Reindexing, Residual-Direction Splitting, and Operator Majorants

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Abstract

The primitive cofactor plane was previously attached to a four-corner coefficient in the first mixed trace of a fixed-shift normalized missing-native Gram. That representation preserves arithmetic signs but does not control a matrix entry or operator norm. We remove this object mismatch. For two active rows $m < n$, every complete common-output key fixes one time j and two squarefree targets

$$mj + h_0 = Q_m S_m, \quad nj + h_0 = Q_n S_n.$$

Writing

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv$$

gives pairwise coprime squarefree coordinates q, u, v , $q \mid |h_0(n - m)|$, and an exact affine solution lattice containing every realized prime key. Without assuming row-shift coprimality its spacing is

$$\text{lcm} \left(\frac{qu}{(m, qu)}, \frac{qv}{(n, qv)} \right);$$

the proved primitive-row property of the inherited physical carrier reduces it to quv . After the inherited unitary row gauge, the target sign at every key is $\mu(u)\mu(v)$.

Let $d_m = ga$, $d_n = gb$, $(a, b) = 1$. We prove that the residual labels are equal if and only if

$$(u, v) = (b, a).$$

Thus the same-residual part is one distinguished primitive direction for each row pair, while every other primitive direction is the erasure part. Defining a literal kernel $L_{mn,q}(u, v)$ by summing $\overline{\beta_{i,m}\beta_{i,n}}/\sqrt{W_m W_n}$ over complete keys with the indicated coordinates, and aggregating the divisor-many q -slices, we obtain the exact formulas

$$\begin{aligned} \tilde{H}_{mn} &= \sum_{(u,v)=1} \mu(u)\mu(v)L_{mn}(u, v), \\ \tilde{R}_{mn} &= \mu(a)\mu(b)L_{mn}(b, a), \quad \tilde{E}_{mn} = \sum_{\substack{(u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u)\mu(v)L_{mn}(u, v). \end{aligned}$$

Hermitian reversal becomes $L_{nm,q}(v, u) = \overline{L_{mn,q}(u, v)}$.

The support-faithful entry kernels are primitive-supported, so their auxiliary diagonal tails vanish and ordinary two-dimensional Abel summation gives exact location-sensitive entry majorants. Collecting these majorants in a symmetric nonnegative matrix P yields $\|H\| \leq \rho(P)$, together with weighted Schur and row-star certificates. These are genuine operator carriers because they retain all entries, but no arithmetic estimate showing that $\rho(P)$ is small is proved. Countermodels separate direct entries from mixed traces, fixed-pair decay from operator control, and general complete-key formulas from direct-cell compressions. No fixed-2, parity-breaking, or prime-pair conclusion is drawn.

Keywords: Möbius function; fixed shift; missing-native Gram; primitive cofactor plane; erasure entry; same-residual direction; two-dimensional Abel summation; Perron majorant.

Convention. Every physical specialization retains one fixed nonzero shift h_0 , literal coefficients, complete native output keys, the actual active support, and raw atomic normalization. No statement is specialized to $h_0 = 2$.

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1 Introduction

The fixed-shift missing-native Gram has the atomic-normalized form

$$G_{\text{nat}} = I + H, \quad H = R + E,$$

where R contains same-residual collisions and E contains unequal-residual collisions created by erasure [2, 3]. The exact source-row gauge of TPC-68 turns every off-diagonal entry into a signed two-affine sum, while preserving the spectrum, entry magnitudes, row stars, and collision graph [3]. Consequently, a direct arithmetic representation of those gauged entries would be a genuine operator interface.

TPC-70 instead begins with the first mixed trace $\text{tr}(ER)$. A same-residual key and an erasure key form a four-corner rectangle. The same-residual key freezes two signs, and the remaining erasure key is reindexed by primitive cofactor coordinates

$$q = (S_m, S_n), \quad S_m = qu, \quad S_n = qv$$

[4]. TPC-71 gives a divisor-lifted Abel certificate for the resulting plane [5]. TPC-73 then observes that the support-faithful zero extension is already primitive-supported; its auxiliary high-diagonal tail is identically zero [6].

That chain is exact but remains attached to a four-corner scalar. Even a perfect estimate for $\text{tr}(ER)$ does not bound E_{mn} , a row star, or $\|H\|$. The present paper returns to the two-corner formula of TPC-68 and applies the cofactor coordinates to the actual common-output key itself. No same-residual anchor is needed to define the coordinates. The row divisors still identify one direction:

$$d_m S_m = d_n S_n \iff (u, v) = (b, a),$$

where $d_m = ga$, $d_n = gb$, and $(a, b) = 1$. The point (b, a) therefore is not deleted from the full plane. It is exactly R_{mn} ; the punctured plane is exactly E_{mn} .

1.1 Main results

The paper proves the following finite, literal statements.

- (1) Every actual complete common-output key determines canonical squarefree, pairwise coprime coordinates q, u, v , with $q \mid |h_0(n - m)|$.
- (2) Fixed (q, u, v) keys lie in an exact affine solution lattice with a general least-common-multiple spacing; the proved primitive-row property of the inherited physical carrier simplifies it to quv . Their gauged Möbius sign is the frozen value $\mu(u)\mu(v)$.
- (3) Same residual is equivalent to the unique primitive direction (b, a) .
- (4) The complete gauged matrices $\tilde{H}, \tilde{R}, \tilde{E}$ have exact entry formulas in the literal kernels $L_{mn,q}$, with normalization $1/\sqrt{W_m W_n}$.
- (5) Reversing the row pair swaps u, v and conjugates the kernel, so the representation is manifestly Hermitian.
- (6) Primitive support removes the auxiliary diagonal tail from every entry kernel. Weighted plaquette costs give explicit entry majorants.
- (7) A symmetric nonnegative matrix of these majorants controls the operator norm through its Perron radius. This is an exact carrier theorem, not a saving for that radius.

1.2 Logical level

The coordinate and matrix identities are L1: they retain the same fixed h_0 , raw masses, complete keys, literal coefficients, actual support, and unitary gauge as the physical Gram. The finite-matrix comparison principles are L0. Estimating the complete growing majorant matrix is the new L2 arithmetic gate. In particular, one fixed pair, one side, a direct-cell compression, or the first mixed trace cannot be substituted for that gate.

1.3 Organization

Section 2 fixes the complete-key native entry. Section 3 constructs its primitive coordinates, and section 4 proves the ray and sign theorems. Section 5 gives the exact $R/E/H$ entry formulas and Hermitian reversal. Section 6 builds entry-to-operator majorants. Countermodels and non-substitution laws appear in section 7. The remaining physical gates and claim ledger are in sections 8 and 9.

2 The complete literal two-corner entry

2.1 Frozen physical carrier

Fix a finite scale X , one nonzero integer shift h_0 , one external block, one direct side, all source and opposite-row blocks, orbit classes, masks, profiles, endpoint restrictions, and the very-high cutoff before any coefficient vector is selected. The opposite direct side is included only by the orthogonal direct sum proved in TPC-68. Let $\mathcal{W}_M$ be the complete set of actual nonzero missing-high atoms in this exact scope, $\mathcal{M}_X$ the declared source-row set, and

$$\mathcal{I}_X = \{i(w) : w \in \mathcal{W}_M\}$$

the set of actual complete native output keys. A key $i(w)$ retains the side, fixed external block, opposite row, γ -coordinate, physical time $j(w)$, and every remaining orthogonal output label.

Every $w \in \mathcal{W}_M$ has a canonical source row $m(w)$, literal coefficient β_w , target, largest prime, cofactor, and residual label

$$N_w = m(w)j(w) + h_0 = Q_w S_w, \quad s(w) = d_{m(w)} S_w. \quad (1)$$

Here β_w is the stripped TPC-68 literal coefficient: it retains all physical shapes, masks, phases, profiles, and terminal weights, but not a second residual Möbius sign. The signed atomic coefficient is $c_w = -\mu(s(w))\beta_w$, as recorded below. The TPC-61 singleton property says that at most one w occurs at a fixed pair (i, m) . Hence define

$$\beta_{i,m} = \begin{cases} \beta_w, & i(w) = i, \ m(w) = m, \\ 0, & \text{if that corner is absent,} \end{cases}$$

and use $N_{i,m}, Q_{i,m}, S_{i,m}, s_{i,m}$ for the corresponding quantities whenever $\beta_{i,m} \neq 0$. More explicitly, the side decomposition is

$$\mathcal{W}_M = \mathcal{W}_M^+ \sqcup \mathcal{W}_M^-, \quad W_m^\sigma = \sum_{\substack{w \in \mathcal{W}_M^\sigma \\ m(w)=m}} |\beta_w|^2, \quad W_m = W_m^+ + W_m^-,$$

where the union is an orthogonal output union, not a cross-side collision set. The raw atomic row mass is

$$W_m = \sum_{\substack{w \in \mathcal{W}_M \\ m(w)=m}} |\beta_w|^2 = \sum_{i \in \mathcal{I}_X} |\beta_{i,m}|^2. \quad (2)$$

No output degree or collision multiplicity divides this mass.

Let

$$\mathcal{M}_M = \{m \in \mathcal{M}_X : W_m > 0\}.$$

Assume $\mathcal{M}_M \neq \emptyset$; otherwise the missing-native analysis is vacuous. All normalized matrices below act on $\ell^2(\mathcal{M}_M)$. Every active target and residual label is on the inherited squarefree carrier:

$$\mu(N_{i,m}), \mu(s_{i,m}) \in \{-1, 1\} \quad (\beta_{i,m} \neq 0). \quad (3)$$

The inherited row divisors satisfy

$$d_m \mid m. \quad (4)$$

The direct coordinate and entry theorems below do not need a coprimality assumption between a source row and h_0 . We therefore prove the general ray lattice first. For sufficiently large X , the actual TPC-61 missing-high carrier additionally satisfies

$$(m, h_0) = 1 \quad (5)$$

for every realized source row [2]; this proved property, rather than a new hypothesis, simplifies the physical ray spacing.

2.2 Normalized missing-native Gram and row gauge

Set

$$c_w = -\mu(s(w))\beta_w, \quad (M_X \zeta)_i = \sum_{\substack{w \in \mathcal{W}_M \\ i(w)=i}} c_w \zeta_{m(w)}, \quad D_M = \text{diag}(W_m).$$

On $S_M = \ell^2(\mathcal{M}_M) = \text{Ran } D_M$, the atomic-normalized missing-native incidence is

$$F = M_X D_M^{\dagger/2} \big|_{S_M}.$$

Its normalized missing-native Gram is

$$G_{\text{nat}} = F^* F = I + H, \quad H = R + E,$$

where all three displayed matrices are Hermitian, H, R, E have zero diagonal, R sums common-output pairs with equal residual labels, and E sums those with unequal residual labels [3].

Define the self-adjoint unitary row gauge

$$U_d = \text{diag}(\mu(d_m))_{m \in \mathcal{M}_M}, \quad \tilde{H} = U_d H U_d, \quad \tilde{R} = U_d R U_d, \quad \tilde{E} = U_d E U_d. \quad (6)$$

For $m \neq n$, put

$$\mathcal{I}_{mn} = \{i \in \mathcal{I}_X : \beta_{i,m} \beta_{i,n} \neq 0\}.$$

The exact TPC-68 two-affine formula is

$$\boxed{\tilde{H}_{mn} = \frac{1}{\sqrt{W_m W_n}} \sum_{i \in \mathcal{I}_{mn}} \mu(N_{i,m}) \mu(N_{i,n}) \overline{\beta_{i,m}} \beta_{i,n}.} \quad (7)$$

Equality of complete keys fixes one common time $j(i)$. Thus (7) is a literal two-corner sum, not a time-only surrogate and not the four-corner coefficient of $\text{tr}(ER)$.

Remark 2.1 (Sides and global row mass). The side is part of the complete output key, so atoms on different sides never form a Gram pair. If $W_m = W_m^+ + W_m^-$, however, the normalization in (7) is still the global W_m . The diagonal contribution of one side alone is $W_m^\pm / W_m$, not automatically 1. Orthogonality across sides therefore does not license a separate identity-diagonal normalization on each side.

2.3 Six-item audit

Every reindexing below retains:

(L1.1) the literal coefficients $\overline{\beta_{i,m}} \beta_{i,n}$;

(L1.2) one fixed nonzero shift h_0 in both targets;

(L1.3) the raw normalization $1/\sqrt{W_m W_n}$;

(L1.4) canonical rows and equality of complete native output keys;

(L1.5) the complete actual squarefree support, including sides, masks, endpoints, and absent-corner zeros; and

(L1.6) zero structural exponent loss for the exact identities, with every later estimate subject to the strict $1/400$ endpoint ledger.

3 Primitive coordinates at one complete output key

For every ordered pair $m \neq n$, define

$$g_{mn} = (d_m, d_n), \quad a_{mn} = \frac{d_m}{g_{mn}}, \quad b_{mn} = \frac{d_n}{g_{mn}}. \quad (8)$$

The three factors g_{mn}, a_{mn}, b_{mn} are squarefree and pairwise coprime, and reversal gives

$$g_{nm} = g_{mn}, \quad a_{nm} = b_{mn}, \quad b_{nm} = a_{mn}.$$

Fix distinct active rows $m < n$ and abbreviate

$$g = g_{mn}, \quad a = a_{mn}, \quad b = b_{mn}.$$

Put

$$\Delta_{mn} = h_0(n - m) \neq 0.$$

Definition 3.1 (Direct-entry cofactor coordinates). For $i \in \mathcal{I}_{mn}$, define

$$q_i = (S_{i,m}, S_{i,n}), \quad S_{i,m} = q_i u_i, \quad S_{i,n} = q_i v_i. \quad (9)$$

Theorem 3.2 (Canonical primitive key coordinates). *At every actual key $i \in \mathcal{I}_{mn}$:*

- (i) q_i, u_i, v_i are squarefree and pairwise coprime;
- (ii) q_i is coprime to ga , u_i is coprime to ga , and v_i is coprime to gb ;
- (iii)

$$\boxed{q_i \mid |\Delta_{mn}|} \quad (10)$$

- (iv) the two largest-prime labels satisfy

$$\boxed{nu_i Q_{i,m} - mv_i Q_{i,n} = \frac{\Delta_{mn}}{q_i}} \quad (11)$$

- (v) the residual labels are equal exactly in one primitive direction:

$$\boxed{s_{i,m} = s_{i,n} \iff (u_i, v_i) = (b, a)}. \quad (12)$$

Proof. The residual labels are

$$s_{i,m} = gaq_i u_i, \quad s_{i,n} = gbq_i v_i.$$

Both are squarefree by (3). Together with $(u_i, v_i) = 1$, which follows from the definition of q_i , this gives the stated squarefreeness and coprimality: squarefreeness of $gaq_i u_i$ makes g, a, q_i, u_i pairwise coprime, and squarefreeness of $gbq_i v_i$ does the same for g, b, q_i, v_i .

The common time in the complete key gives

$$\begin{aligned} nN_{i,m} - mN_{i,n} &= n(mj(i) + h_0) - m(nj(i) + h_0) \\ &= h_0(n - m) = \Delta_{mn}. \end{aligned}$$

Substitute $N_{i,m} = Q_{i,m}q_iu_i$ and $N_{i,n} = Q_{i,n}q_iv_i$. This yields

$$q_i(nu_iQ_{i,m} - mv_iQ_{i,n}) = \Delta_{mn},$$

proving (10) and (11).

Finally,

$$s_{i,m} = s_{i,n} \iff au_i = bv_i.$$

If $au_i = bv_i$, then $(a, b) = 1$ gives $u_i = bk$, $v_i = al$. The equality gives $k = \ell$, while $(u_i, v_i) = 1$ forces $k = \ell = 1$. The converse is immediate. $\square$

Remark 3.3 (No anchor is used). The coordinate q_i is recovered from the two cofactors at the same complete output key. A same-residual return key is not part of [definition 3.1](#). The row-divisor split only marks which point of the full primitive plane belongs to R .

Corollary 3.4 (Divisor-many content slices). *For fixed m, n, h_0 , every actual entry key belongs to one of at most $\tau(|\Delta_{mn}|)$ content slices. This divisor bound controls multiplicity only; it gives no cancellation among literal slice amplitudes.*

Remark 3.5 (Actual support is smaller). The arithmetic conditions above are necessary, not sufficient. Primality of $Q_{i,m}, Q_{i,n}$, their largest-prime status, the physical interval, complete-key incidence, sides, masks, profiles, and endpoints may delete points from a primitive direction. They are retained inside the literal kernel defined below.

4 Affine rays and the direct-entry sign

4.1 Exact ray spacing

Theorem 4.1 (General fixed- (q, u, v) direct-entry ray). *Fix m, n, h_0 , and suppose one actual key realizes coordinates (q, u, v) , with data*

$$(j_0, Q_{m,0}, Q_{n,0})$$

exists, define

$$\lambda_{mn}(q, u, v) = \text{lcm}\left(\frac{qu}{(m, qu)}, \frac{qv}{(n, qv)}\right) \quad (13)$$

Then every other physical key with the same (q, u, v) has a unique integer z for which

$$j = j_0 + \lambda_{mn}(q, u, v)z, \quad (14)$$

$$Q_m = Q_{m,0} + \frac{m\lambda_{mn}(q, u, v)}{qu}z, \quad (15)$$

$$Q_n = Q_{n,0} + \frac{n\lambda_{mn}(q, u, v)}{qv}z \quad (16)$$

Proof. For two keys with the same coordinates, take differences. From

$$mj + h_0 = quQ_m, \quad nj + h_0 = qvQ_n$$

one obtains

$$m\delta j = qu\delta Q_m, \quad n\delta j = qv\delta Q_n.$$

The first divisibility is equivalent to $qu/(m, qu) \mid \delta j$, and the second to $qv/(n, qv) \mid \delta j$. Their simultaneous solution lattice is therefore exactly $\delta j = \lambda_{mn}(q, u, v)z$. Substitution gives the two displayed prime increments; they are integers by the definition of λ_{mn} . $\square$

Remark 4.2 (Containment, not prime production). The lattice theorem is one-way. It places every realized key on the displayed lattice; it does not assert that every integer z produces prime largest factors or survives the largest-prime test, masks, complete output labels, physical interval, or endpoints.

Corollary 4.3 (Physical primitive-row spacing). *On the inherited physical carrier*

$$(mn, h_0) = 1,$$

one has

$$(m, qu) = (n, qv) = 1, \quad \lambda_{mn}(q, u, v) = quv.$$

Consequently,

$$j = j_0 + quvz, \quad Q_m = Q_{m,0} + mvz, \quad Q_n = Q_{n,0} + nu z.$$

Proof. Apply (5) to both realized rows. Since $qu \mid mj + h_0$, any common divisor of m and qu divides h_0 ; hence it is 1. The argument for n, qv is identical. Now q, u, v are pairwise coprime, so $\text{lcm}(qu, qv) = quv$. Substitute this value into equations (14) to (16). $\square$

For a complete key i , let $\ell(i)$ denote the tuple of all its labels other than the time coordinate. This tuple retains, in particular, the side, external block, opposite row, γ -coordinate, orbit class, and every remaining orthogonal label.

Corollary 4.4 (Ray census). *If the allowed time interval has length J , then*

$$\#\{i \in \mathcal{I}_{mn} : (q_i, u_i, v_i) = (q, u, v), \ell(i) = \ell_0\} \leq 1 + \left\lfloor \frac{J}{\lambda_{mn}(q, u, v)} \right\rfloor$$

for each fixed label tuple ℓ_0 . The unrestricted key count requires summation over the realized ℓ_0 's; no label multiplicity is hidden in the displayed bound, and the sum does not exploit literal phases.

4.2 Frozen target sign

Theorem 4.5 (Primitive sign of one direct key). *For every $i \in \mathcal{I}_{mn}$,*

$$\boxed{\mu(N_{i,m})\mu(N_{i,n}) = \mu(u_i)\mu(v_i)}. \quad (17)$$

Consequently, the Möbius sign is constant along every arithmetic ray in theorem 4.1, irrespective of whether its spacing simplifies to quv .

Proof. Each largest prime is coprime to its cofactor on the squarefree target support, so

$$\mu(N_{i,m}) = -\mu(S_{i,m}), \quad \mu(N_{i,n}) = -\mu(S_{i,n}).$$

The two minus signs cancel. The pairwise coprimality from theorem 3.2 gives

$$\mu(q_i u_i)\mu(q_i v_i) = \mu(q_i)^2 \mu(u_i)\mu(v_i) = \mu(u_i)\mu(v_i).$$

$\square$

Corollary 4.6 (Same-residual sign check). *At the distinguished point $(u, v) = (b, a)$,*

$$\mu(u)\mu(v) = \mu(a)\mu(b) = \mu(d_m)\mu(d_n).$$

This agrees with the same-residual sign-freezing theorem of TPC-68.

Proof. The first equality is symmetry. Since the squarefree factors g, a, b are pairwise coprime,

$$\mu(d_m)\mu(d_n) = \mu(ga)\mu(gb) = \mu(g)^2 \mu(a)\mu(b).$$

$\square$

Remark 4.7 (What can oscillate). The theorem freezes only the target Möbius sign inside a fixed ray. The literal coefficient $\overline{\beta_{i,m}}\beta_{i,n}$ may carry complex phases and amplitude variation, and the set of surviving prime points may be highly irregular.

5 Exact direct-entry primitive-plane representation

5.1 Literal pair kernels

Definition 5.1 (Complete-key entry kernel). For $m \neq n$, $q, u, v \geq 1$, define

$$L_{mn,q}(u, v) = \frac{1}{\sqrt{W_m W_n}} \sum_{\substack{i \in \mathcal{I}_{mn} \\ (q_i, u_i, v_i) = (q, u, v)}} \overline{\beta_{i,m}} \beta_{i,n}. \quad (18)$$

The sum retains every complete-key label; it is zero when the indicated actual fiber is empty. Aggregate contents before absolute values:

$$L_{mn}(u, v) = \sum_{q \mid |\Delta_{mn}|} L_{mn,q}(u, v), \quad (u, v) = 1. \quad (19)$$

Its support-faithful zero extension to $\mathbb{N}^2$ is

$$L_{mn}^0(u, v) = \begin{cases} L_{mn}(u, v), & (u, v) = 1, \\ 0, & \gcd(u, v) > 1. \end{cases} \quad (20)$$

Different q -slices may cancel at the same primitive coordinate, so only

$$\text{supp } L_{mn}^0 \subseteq \bigcup_{q \mid |\Delta_{mn}|} \text{supp } L_{mn,q}$$

is asserted.

For each ordered pair $m \neq n$, define the punctured erasure kernel

$$L_{mn}^E(u, v) = L_{mn}^0(u, v) - L_{mn}(b_{mn}, a_{mn}) e_{b_{mn}, a_{mn}}(u, v), \quad e_{r,s}(u, v) = \mathbf{1}_{(u,v)=(r,s)}. \quad (21)$$

Theorem 5.2 (Exact two-corner primitive planes). *For every pair of distinct active rows $m < n$,*

$$\boxed{\tilde{H}_{mn} = \sum_{u,v \geq 1} \mu(u) \mu(v) L_{mn}^0(u, v),} \quad (22)$$

$$\boxed{\tilde{R}_{mn} = \mu(a) \mu(b) L_{mn}(b, a),} \quad (23)$$

$$\boxed{\tilde{E}_{mn} = \sum_{u,v \geq 1} \mu(u) \mu(v) L_{mn}^E(u, v) = \sum_{\substack{(u,v)=1 \\ (u,v) \neq (b,a)}} \mu(u) \mu(v) L_{mn}(u, v).} \quad (24)$$

The normalizing factor in every kernel is $1/\sqrt{W_m W_n}$.

Proof. Partition the complete finite sum (7) by its canonical triple (q_i, u_i, v_i) . By [theorem 3.2](#), every q_i divides $|\Delta_{mn}|$ and every (u_i, v_i) is primitive. By [theorem 4.5](#), the sign on that fiber is $\mu(u_i) \mu(v_i)$. This proves (22).

The definition of R retains exactly the keys with $s_{i,m} = s_{i,n}$. By (12), these are exactly the keys at $(u_i, v_i) = (b, a)$, and their sign is $\mu(b) \mu(a)$. This proves (23). The complementary keys form E , giving (24). $\square$

Proposition 5.3 (Atomic-mass audit). *For every distinct active pair,*

$$\sum_{q, u, v \geq 1} |L_{mn,q}(u, v)| \leq \frac{\sum_{i \in \mathcal{I}_{mn}} |\beta_{i,m} \beta_{i,n}|}{\sqrt{W_m W_n}} \leq 1. \quad (25)$$

In particular,

$$|\tilde{R}_{mn}| + |\tilde{E}_{mn}| \leq 1, \quad |\tilde{H}_{mn}| \leq 1.$$

These are entry-normalization checks, not operator-norm savings.

Proof. The canonical triples partition $\mathcal{I}_{mn}$. Apply the triangle inequality inside each fiber and then Cauchy–Schwarz to the common-key subset:

$$\sum_{i \in \mathcal{I}_{mn}} |\beta_{i,m} \beta_{i,n}| \leq \left(\sum_i |\beta_{i,m}|^2 \right)^{1/2} \left(\sum_i |\beta_{i,n}|^2 \right)^{1/2} = \sqrt{W_m W_n}.$$

The R - and E -keys are disjoint subfamilies of the same partition. $\square$

Remark 5.4 (One point is not one key). The value $L_{mn}(b, a)$ may aggregate many contents, times, sides, γ -coordinates, and complete keys. The theorem identifies one primitive direction, not one physical atom.

Remark 5.5 (Two-corner versus four-corner normalization). The direct kernel uses $1/\sqrt{W_m W_n}$. The TPC–70 mixed-trace rectangle uses four coefficients and $1/(W_m W_n)$. Replacing one by the other changes both the object and its physical scale.

5.2 Hermitian reversal

Theorem 5.6 (Reversed-pair conjugacy). *For every $m \neq n$, $q, u, v \geq 1$,*

$$\boxed{L_{nm,q}(v, u) = \overline{L_{mn,q}(u, v)}, \quad L_{nm}^0(v, u) = \overline{L_{mn}^0(u, v)}}. \quad (26)$$

For the punctured kernels,

$$L_{nm}^E(v, u) = \overline{L_{mn}^E(u, v)}.$$

Consequently, (22)–(24) are manifestly Hermitian.

Proof. The complete-key set is the same: $\mathcal{I}_{nm} = \mathcal{I}_{mn}$. Reversing the rows swaps the two cofactors, hence (q, u, v) becomes (q, v, u) . It also replaces

$$\frac{\overline{\beta_{i,m}} \beta_{i,n}}{\sqrt{W_m W_n}} \quad \text{by} \quad \frac{\overline{\beta_{i,n}} \beta_{i,m}}{\sqrt{W_n W_m}},$$

which is its complex conjugate. Summing gives the first identity. The indexed row-divisor factors swap under reversal, so the ordered puncture (b_{mn}, a_{mn}) becomes $(b_{nm}, a_{nm}) = (a_{mn}, b_{mn})$. The remaining assertions follow. $\square$

5.3 Direct-cell compression is optional

Under the identical-support direct-cell specialization of TPC–68, the γ -sum in (18) may be compressed into the literal overlap

$$\Gamma_{m,n}(j) = \sum_{\gamma} K_{m\gamma} K_{n\gamma} |\psi_{\gamma}(j)|^2.$$

Here $K_{m\gamma} \in \{0, 1\}$ is the inherited static support mask and $\psi_{\gamma}(j)$ is the inherited opposite-row profile and output phase. The notation ψ_{γ} avoids confusion with the cofactor content q . Outside that declared specialization, the masks and first-failure conditions may depend on γ . Then the complete-key formula (18), rather than a compressed $\Gamma_{m,n}$, is the theorem. No later result requires identical support.

6 From primitive entry kernels to operator majorants

6.1 Tail-free entrywise Abel identity

Set

$$M(x) = \sum_{r \leq x} \mu(r)$$

and, for a zero-extended finite kernel F ,

$$F^{[d]}(x, y) = F(dx, dy), \quad \Delta_{12} F(x, y) = F(x, y) - F(x+1, y) - F(x, y+1) + F(x+1, y+1).$$

Define the location-sensitive weighted plaquette cost

$$\mathcal{C}_\square(F) = \sum_{x,y \geq 1} |M(x)M(y)| |\Delta_{12}F(x,y)|. \quad (27)$$

For comparison with the divisor lift of TPC-73, also define its exact large-diagonal tail

$$\mathcal{T}_D(F) = \sum_{d > D} |\mu(d)| \sum_{x,y \geq 1} |\mu(dx)\mu(dy)F(dx,dy)|. \quad (28)$$

Theorem 6.1 (Primitive entry Abel certificate). *For every $m \neq n$,*

$$\tilde{H}_{mn} = \sum_{x,y \geq 1} M(x)M(y)\Delta_{12}L_{mn}^0(x,y), \quad (29)$$

$$\tilde{E}_{mn} = \sum_{x,y \geq 1} M(x)M(y)\Delta_{12}L_{mn}^E(x,y). \quad (30)$$

Consequently,

$$|\tilde{H}_{mn}| \leq \mathcal{C}_\square(L_{mn}^0), \quad (31)$$

$$|\tilde{E}_{mn}| \leq \mathcal{C}_\square(L_{mn}^E), \quad (32)$$

$$|\tilde{R}_{mn}| = |L_{mn}(b_{mn}, a_{mn})|. \quad (33)$$

Moreover, for $d \geq 2$,

$$(L_{mn}^0)^{[d]} \equiv 0, \quad (L_{mn}^E)^{[d]} \equiv 0.$$

Thus the auxiliary diagonal tails of the support-faithful entry kernels vanish for every cutoff $D \geq 1$.

Proof. Both kernels are finitely supported on the primitive lattice. Apply two-dimensional summation by parts to (22) and (24), then take absolute values. If $d \geq 2$, the point (dx, dy) has greatest common divisor at least d , so both primitive-supported kernels vanish there. The residual magnitude follows from (23). $\square$

Remark 6.2 (No cancellation theorem yet). Equations (29) and (30) are exact. They do not assert that either weighted plaquette cost is smaller than the literal entry mass. The prime number theorem statement $M(x) = o(x)$ does not by itself supply a uniform fixed-power saving over the growing row-pair family.

6.2 The nonnegative entry matrix

For $m < n$, define

$$P_{mn}^H = \mathcal{C}_\square(L_{mn}^0), \quad (34)$$

$$P_{mn}^E = \mathcal{C}_\square(L_{mn}^E), \quad (35)$$

$$P_{mn}^R = |L_{mn}(b_{mn}, a_{mn})|. \quad (36)$$

Set

$$P_{nm}^\chi = P_{mn}^\chi, \quad P_{mm}^\chi = 0 \quad (\chi \in \{H, E, R\}).$$

This upper-triangle definition makes all three matrices symmetric without an orientation convention. It agrees with the reversed kernels by theorem 5.6, since transposition and conjugation preserve $\mathcal{C}_\square$, while $(b_{nm}, a_{nm}) = (a_{mn}, b_{mn})$.

Theorem 6.3 (Literal Perron operator certificates). *The exact physical matrices satisfy*

$$\boxed{\|H\| = \|\tilde{H}\| \leq \|P^H\| = \rho(P^H),} \quad (37)$$

$$\boxed{\|E\| = \|\tilde{E}\| \leq \|P^E\| = \rho(P^E),} \quad (38)$$

$$\boxed{\|R\| = \|\tilde{R}\| \leq \|P^R\| = \rho(P^R).} \quad (39)$$

For every $c = (c_m) \in (0, \infty)^{\mathcal{M}_M}$,

$$\rho(P^H) \leq \max_m \frac{\sum_n P_{mn}^H c_n}{c_m}, \quad \rho(P^H) = \inf_{c_m > 0} \max_m \frac{\sum_n P_{mn}^H c_n}{c_m}. \quad (40)$$

The analogous statements hold for P^E and P^R .

Proof. The entry inequalities in [theorem 6.1](#) give $|\tilde{H}_{mn}| \leq P_{mn}^H$, and similarly for E, R . For arbitrary vectors x, y ,

$$|\langle x, \tilde{H}y \rangle| \leq \langle |x|, P^H |y| \rangle \leq \|P^H\| \|x\|_2 \|y\|_2.$$

Hence $\|\tilde{H}\| \leq \|P^H\|$. The majorant is symmetric and nonnegative, so its operator norm is its Perron root. The weighted bound and the reducible-safe infimum formula are the Collatz–Wielandt characterization [\[1\]](#). The other matrices are identical. $\square$

Corollary 6.4 (Schur and row-star forms). *Let*

$$\Delta_H = \max_m \#\{n : P_{mn}^H > 0\}.$$

Then

$$\|H\| \leq \max_m \sum_n P_{mn}^H \leq \sqrt{\Delta_H} \max_m \left(\sum_n (P_{mn}^H)^2 \right)^{1/2}. \quad (41)$$

The weighted form (40) may be strictly sharper than the unweighted maximum row sum.

Corollary 6.5 (A conditional normalized-Gram consequence). *For a family indexed by X , if a literal certificate proves*

$$\rho(P_X^H) \leq \eta_X < 1,$$

then

$$1 - \eta_X \leq \lambda_{\min}(G_{\text{nat}, X}) \leq \lambda_{\max}(G_{\text{nat}, X}) \leq 1 + \eta_X.$$

This implication is deterministic. The hypothesis is not proved here.

Proof. Use $G_{\text{nat}} = I + H$ and [\(37\)](#). $\square$

6.3 Completion freedom returns at the estimate level

The physical entry samples only primitive sites. Therefore the analytic completion gauge of TPC–73 can be applied separately to every L_{mn}^0 . A nonprimitive completion may reduce its $d = 1$ variation, but all higher diagonal slices and the exact tail of that same completion must be restored. To obtain an operator theorem, the complete family of completion rules must be declared before the later coefficient extremizer is chosen, must respect [\(26\)](#), and must produce one controlled majorant matrix. Pairwise optimization with no uniform summability is not an L2 result.

7 Countermodels and non-substitution laws

7.1 A literal orphan erasure edge

Take the finite fixed-shift interface

$$m = 2, \quad n = 3, \quad h_0 = 1, \quad d_m = d_n = 1,$$

with one complete common-output key at time $j = 3$. Its targets are

$$2j + 1 = 7 = 7 \cdot 1, \quad 3j + 1 = 10 = 5 \cdot 2.$$

Thus

$$q = 1, \quad (u, v) = (1, 2), \quad (b, a) = (1, 1).$$

The residuals 1 and 2 are unequal. Choose one literal coefficient at each row, both equal to 1; then $W_m = W_n = 1$. With a finite high threshold below 5, the largest-prime labels are literal and all target and residual Möbius values are nonzero. The exact matrices are

$$R = 0, \quad E = H = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Indeed,

$$L_{mn,1}(1,2) = 1, \quad \mu(1)\mu(2) = -1.$$

Every four-corner ER anchor coefficient vanishes because $R = 0$, whereas the direct erasure entry and its operator norm are nonzero. This is a literal finite-interface witness that the anchored primitive plane cannot substitute for the direct plane. It does not assert that orphan edges carry nonnegligible mass in the full asymptotic carrier.

7.2 A literal same-residual direction

Take

$$m = 2, \quad n = 3, \quad h_0 = 1, \quad d_m = 2, \quad d_n = 3,$$

so $g = 1$, $a = 2$, and $b = 3$. At time $j = 7$,

$$2j + 1 = 15 = 5 \cdot 3, \quad 3j + 1 = 22 = 11 \cdot 2.$$

Thus

$$q = 1, \quad (u, v) = (3, 2) = (b, a),$$

and the two residual labels agree:

$$d_m S_m = 2 \cdot 3 = 6 = 3 \cdot 2 = d_n S_n.$$

This finite arithmetic key checks the distinguished R -direction. Entry into a particular asymptotic physical family still requires all of that family's literal masks and endpoint conditions.

7.3 The same-residual point need not cancel

Let the only nonzero direction for one row pair be

$$L_{mn}^0 = ce_{b,a}.$$

Then

$$\tilde{R}_{mn} = \mu(a)\mu(b)c, \quad \tilde{E}_{mn} = 0.$$

The full magnitude is $|c|$. Primitive support, affine-ray structure, and the divisor bound on q do not create Möbius cancellation inside the distinguished direction. Literal phases across keys are a separate possible mechanism.

7.4 One common key does not have unit normalized weight

Let each of two rows carry T literal unit coefficients, but let only one complete output key be common to the rows. Then

$$W_m = W_n = T$$

and the common key contributes

$$\frac{\overline{\beta_{i,m}}\beta_{i,n}}{\sqrt{W_m W_n}} = \frac{1}{T},$$

not 1. Neither the singleton property nor equality of the common key permits replacing the raw row masses by collision degrees. This also illustrates why the entry audit (25) is a normalization bound rather than a uniform saving.

7.5 Mixed-trace cancellation does not control an entry

Consider the abstract Hermitian zero-diagonal matrices

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}.$$

Then $ER \neq 0$, but

$$\text{tr}(ER) = 0, \quad \|E\| = \|R\| = 1.$$

Thus even cancellation of a nonzero first mixed carrier does not bound either factor. The direct-entry theorem is a new object interface, not a corollary of a scalar trace estimate.

7.6 Fixed-pair decay does not control the operator

For $N \geq 2$, let A_N be the N -by- N matrix with zero diagonal and every off-diagonal entry equal to $1/(N-1)$. For each fixed pair,

$$(A_N)_{mn} \longrightarrow 0,$$

while

$$\|A_N\| = 1$$

because the all-ones vector is an eigenvector. Hence fixed-pair little-oh notation for $\mathcal{C}_\square(L_{mn}^0)$ is insufficient. A row, weighted Schur, or Perron estimate over the complete growing pair family is necessary.

7.7 Pre-aggregation costs can lose literal cancellation

In the ambient finite-kernel class, two content slices F and $-F$ have

$$L_{mn} = F + (-F) = 0,$$

although their separate weighted costs are positive. Therefore

$$\mathcal{C}_\square\left(\sum_q L_{mn,q}\right) \leq \sum_q \mathcal{C}_\square(L_{mn,q})$$

is valid but may be strict. This model does not assert that exact cancellation occurs in every physical family; it fixes the legal order of operations.

7.8 A time-only compression can change the carrier

Equality of complete native outputs implies equality of time, but the converse fails. Two atoms at the same time with different side, γ , block, or remaining output labels do not form a Gram collision. Replacing $\mathcal{I}_{mn}$ in (18) by a time set may therefore create spurious two-corner products. The compressed direct-cell $\Gamma_{m,n}(j)$ formula is legal only under its identical-support hypothesis.

7.9 Failure of the majorant is not failure of cancellation

A large value of $\rho(P^H)$ proves only that the present absolute-value certificate is inconclusive. The signed matrix H may still be small because of cancellations between plaquettes, row pairs, contents, sides, or literal phases discarded by P^H . To close the Abel–Perron route negatively, one needs an actual-carrier lower obstruction for every declared admissible completion or for the exact majorant architecture, at a nonnegligible physical mass scale.

8 The new physical gate and the adaptive next door

Let $X \rightarrow \infty$ through one declared physical family. All active row sets, complete keys, kernels, and majorants now depend on X ; the index is suppressed unless needed. Every asymptotic statement below uses the same fixed h_0 , literal coefficient class, raw atomic normalization, and actual support.

8.1 What the exact carrier has closed

TPC-73 left open whether a primitive plane could be attached to a two-corner matrix entry. [Theorem 5.2](#) answers this affirmatively:

$$\text{every off-diagonal entry} \longleftrightarrow \text{one literal primitive plane.}$$

For each ordered row pair, the same-residual entry is its pair-dependent distinguished direction (b_{mn}, a_{mn}) , and the erasure entry is the primitive complement. Consequently, no anchor-observability hypothesis is needed to see an erasure edge, and orphan edges are included automatically.

This closes an L1 object mismatch. It does not close any of the following L2 estimates.

8.2 Four remaining operator gates

Gate A: mass-normalized plaquette or completion control. One must estimate the actual kernels

$$\mathcal{C}_{\square}(L_{mn}^0)$$

or the full divisor-variation cost of one declared nonprimitive completion. Smooth envelopes, time-only kernels, and support counts do not replace the literal complex amplitudes.

Gate B: complete pair aggregation. The desired conclusion must control the complete matrix, for example

$$\rho(P_X^H) \leq \eta_X$$

at the declared physical scale. Fixed-pair cancellation, density-one rows, or one side do not imply this estimate. Weighted Perron vectors are allowed only when fixed before the later coefficient extremizer.

Gate C: the same-residual sector. The full Gram requires $H = R + E$. The target Möbius sign of R is frozen. Any saving there must come from literal phases, sparsity, geometry, or a separate deterministic mass bound. An estimate for E alone is not an estimate for H .

Gate D: physical reassembly. Even a complete operator estimate must be combined with terminal activation, a represented lower frame, missing-mass comparison, nuisance shorting, geometric endpoints, and final packet boundaries on the same carrier and coefficient scope.

8.3 Exact stop rules

- (G1) If the canonical P_X^H is too large by a proved lower obstruction, only the canonical zero-extension Abel-Perron certificate is closed.
- (G2) To close completion-based entry certificates, the obstruction must cover every admissible completion rule or the corresponding completion infimum, with Hermitian and uniform-quantifier constraints retained.
- (G3) If orphan edges carry nonnegligible operator mass, that closes an anchor-only observability route but not the direct-entry route.
- (G4) If the same-residual majorant is too large, erasure cancellation cannot repair the triangle estimate automatically; one may need a signed full- H moment rather than separate R/E norms.
- (G5) If the actual total fixed-power loss reaches $1/400$, including equality, the current endpoint implication is closed.
- (G6) None of these outcomes is a theorem about the special shift $h_0 = 2$, the sieve parity barrier, prime pairs, or twin primes.

8.4 TPC-75 decision

The direct carrier changes the priority of the next plane paper. TPC-75 will compare two coefficient-faithful entry certificates:

- (i) the canonical primitive zero extension, whose auxiliary diagonal tail is zero; and
- (ii) a q -compatible nonprimitive completion that moves common coordinate factors into existing literal content slices.

The target is a complete physical census of weighted plaquettes, large-content mass, row stars, and the resulting Perron matrix. A positive result must bound the actual growing matrix. A negative result must exhibit a nonnegligible actual-carrier obstruction, not merely a rough ambient upper bound. TPC-76 will then quantify what four-corner anchors observe and isolate the orphan complement, providing a hybrid fallback rather than replacing the direct representation.

9 Claim ledger and conclusion

9.1 Exact conclusions

The paper proves:

- (1) canonical primitive coordinates for every actual complete two-corner key;
- (2) the divisor condition $q \mid |h_0(n - m)|$, the exact linear Diophantine identity, and the general least-common-multiple solution lattice, with quv on the inherited physical carrier;
- (3) the frozen direct-entry sign $\mu(u)\mu(v)$;
- (4) equivalence between equal residuals and the single direction $(u, v) = (b, a)$;
- (5) exact literal formulas for every entry of $\tilde{H}, \tilde{R}, \tilde{E}$;
- (6) reversed-pair conjugacy and therefore Hermitian compatibility;
- (7) tail-free Abel identities for the support-faithful entry kernels; and
- (8) entrywise, Schur, row-star, weighted Collatz, and Perron operator certificates for the complete literal matrix.

9.2 Statements not proved

The paper does not prove:

- that a physical entry has cancellation relative to its literal mass;
- that $\mathcal{C}_{\square}(L_{mn}^0)$ is small uniformly in a growing row-pair family;
- that $\rho(P_X^H)$, $\rho(P_X^E)$, or $\rho(P_X^R)$ has any desired asymptotic saving;
- a uniformly controlled low-cost analytic completion;
- cancellation in the same-residual sector from target Möbius signs;
- closure of terminal activation, represented lower frame, missing-mass comparison, nuisance shorting, endpoints, or final boundary reconstruction;
- the strict $1/400$ endpoint inequality for one complete physical chain;
- a statement special to $h_0 = 2$; or
- a parity-breaking, prime-pair, or twin-prime theorem.

9.3 Audit levels

Result	Scope	Level
General least-common-multiple ray and matrix comparison	Finite arithmetic triples and finite matrices	L0
Physical quv ray specialization	Sufficiently large TPC-61 missing-high carrier	L1
Direct-entry primitive representation	Complete literal fixed- h_0 missing-high carrier	L1
Same-residual direction and Hermitian reversal	Every actual active row pair	L1
Mass-normalized entry saving	Complete growing literal key family	Open L2
Perron/row-star saving	Complete growing active-row matrix	Open L2
Physical reassembly	One common carrier and coefficient scope	Open

9.4 Conclusion

The primitive cofactor plane is no longer confined to a four-corner mixed trace. It represents every literal off-diagonal entry of the normalized missing-native Gram directly:

$$\tilde{H}_{mn} = \sum_{(u,v)=1} \mu(u)\mu(v)L_{mn}(u,v).$$

For each ordered row pair, the same-residual entry is its unique pair-dependent direction (b_{mn}, a_{mn}) ; the erasure entry is the remaining primitive plane. This is a genuine L1 operator-carrier advance because no entry, complete key, literal coefficient, or raw normalization is discarded.

The hard question has consequently become sharper, not solved. One must control the weighted plaquette or completion cost for all actual row pairs and then prove that the resulting Perron matrix is small at the physical scale. TPC-75 will test that certificate on the literal carrier. If it fails by an actual obstruction, the fallback is a signed moment or hybrid direct/anchor architecture, not a claim that arithmetic cancellation itself has failed.

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